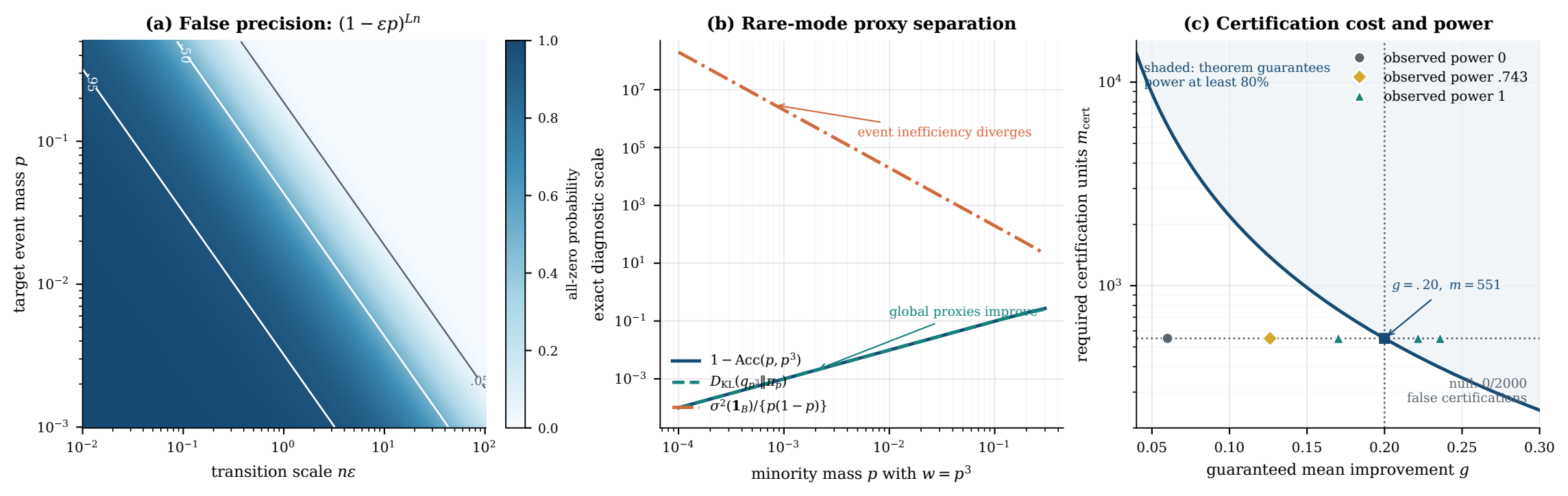

(a) False precision: $(1-\varepsilon p)^{Ln}$
target event mass p
transition scale $n\varepsilon$
all-zero probability
.95
.50
.05
(b) Rare-mode proxy separation
exact diagnostic scale
minority mass p with $w = p^3$
event inefficiency diverges
global proxies improve
$1 - \mathrm{Acc}(p, p^3)$
$D_{\mathrm{KL}}(q_{p^3} \| \pi_p)$
$\sigma^2(\mathbf{1}_B)/\{p(1-p)\}$
(c) Certification cost and power
required certification units m_{cert}
guaranteed mean improvement g
shaded: theorem guarantees
power at least 80%
observed power 0
observed power .743
observed power 1
$g = .20,\ m = 551$
null: 0/2000
false certifications